MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5641

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Subminiature, high vacuum, half-wave rectifier

Outline --- 3-3 (EIA)

Base --- E8-10

Envelope --- T3

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	nc	a	h	a	k	h	nc	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Epp	epx	Ehk	ib	Io	i(surge)	R1	C1	TE	Alt
Unit:	V	V ac	v	v	ma	mA dc	a	Ohms	µF	°C	ft
Maximum:	6.6	---	775	465	300	50	1.1	---	---	220	Note 1
Minimum:	6.0	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	275	---	0	---	---	---	5,000	16	---	---

GENERAL:

First article inspection - Required (see note 2)

FSC 5960

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:00Z
Check the source to verify that this is the current version before use.

Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Insp level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 1								
1301	Heater current			0.4	II	If	420	480	mA
1353	Operation of rectifiers	3,4		0.4	II	Io	47	---	mA dc
1336	Heater-cathode leakage		Ehk = 465 V dc	0.4	II	Ihk	---	50	µA dc
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes		E (a to all) = -300 V dc	2.5	I	R	50	---	Meg
1031	Variable-frequency vibration		No voltages	10.0	See note 5	---	---	---	---
---	Variable-frequency vibration - test end points								
1336	Heater-cathode leakage		Ehk = 465 V dc	---	---	Ihk	---	100	µA dc
1353	Operation of rectifiers	3		---	---	Io	45	---	mA dc
1231	Emission		Eb = 30 V dc	2.5	I	Is	100	---	mA dc
1116	Lead fatigue			2.5	Code F	---	---	---	---
2126	Envelope strain			2.5	I	---	---	---	---
1041	Shock		450 G; Ehk = +100 V dc; Epp = 0	6.5	See note 5	---	---	---	---
1031	Vibration-fatigue			6.5	See note 5	---	---	---	---
---	Shock and vibration-fatigue-test end points								
1336	Heater-cathode leakage		Ehk = 465 V dc	---	---	Ihk	---	100	µA dc
1353	Operation of rectifiers	3		---	---	Io	45	---	mA dc
1105	Permanence of marking			---	---	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:00Z
Check the source to verify that this is the current version before use.

Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.0 V; 1 min "on", 4 min "off"; Ehk = 140 V ac; Eb = 0	---	---	---	---
1501	Intermittent life	6,7	TE = +200°C	---	---	---	---
---	Intermittent life-test end points (500 hours)	8					
1301	Heater current			If	414	492	mA
1336	Heater-cathode leakage		Ehk = 465 V dc	Ihk	---	60	µA dc
1353	Operation of rectifiers	3		Io	43	---	mA dc
1353	Change in operation of rectifiers of individual tubes	3		ΔIo t	---	15	%
1211	Insulation of electrodes		E (a to all) = -300 V dc	R	25	---	Meg
---	Intermittent life-test end points (1,000 hours)	8					
1301	Heater current			If	414	492	mA
1336	Heater-cathode leakage		Ehk = 465 V dc	Ihk	---	100	µA dc
1353	Operation of rectifiers	3		Io	43	---	mA dc
1353	Change in operation of rectifiers of individual tubes	3		ΔIo t	---	15	%
1211	Insulation of electrodes		E (a to all) = -300 V dc	R	25	---	Meg

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. First article sample inspection shall conform to the requirements of appendix F of MIL-E-1 and shall consist of performing all tests specified in this TSS. Invitation for bids should provide that the preparing activity reserves the right to waive the requirements for first article samples for those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.
3. In a half-wave circuit, adjust Zp so that Io = 50 mA dc with a tube having an Etd of 25.0 V dc at Is = 100 mA dc. Ehk = Eo + 117 V ac.
4. This test shall be performed at the conclusion of the holding period.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:00Z
Check the source to verify that this is the current version before use.

5. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code E, shall apply.

6. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey tube drop of Etd = 25.0 V dc (±10 percent) at Is = 100 mA dc per anode, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

7. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples and shall not exceed a total of three tube failures.

8. The allowable defectives per characteristic shall not be greater than 1 for the first sample and shall not be greater than 3 for combined samples.

9. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 99
DLA - ES

User activities:
Army - AR
Navy - AS, MC, CG, SH
Air Force - 11, 19

Agent:
DLA - ES

Preparing activity:
Air Force - 85

(Project 5960-3294)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:00Z
Check the source to verify that this is the current version before use.